SAINT MICHAEL COLLEGE OF CARAGA

PREDICTING STUDENT ACADEMIC STANDING USING MACHINE LEARNING MODELS

WITH EXPLAINABLE AI

A Thesis Presented To

The Faculty of

College of Computing and Information Sciences

In Partial Requirements for the Degree

BACHELOR OF SCIENCE IN COMPUTER SCIENCE



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This thesis entitled **"PREDICTING STUDENT ACADEMIC STANDING USING RANDOM FOREST ALGORITHM"** presented and submitted by Rexelle V. Azarraga, Diether F. Flores, Trixia Lee C. Lope, Kevin Jun A. Maestro, Janryl S. Signar in partial fulfillment of the requirements for the degree of College of Computing in Computer Science is hereby accepted and recommended for Oral Examination.

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ABSTRACT

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(RESEARCH DESIGN)

I. OBJECTIVES

II. METHODOLOGY

III. FINDINGS

IV. RECOMMENDATIONS

KEYWORDS:

DEDICATION

This thesis is addressed to our beloved families, who have given us so much love, strength, and motivation throughout our journey.

To our parents, who have sacrificed so much for us to continue our studies, and to whom we've never stopped trying to live up to their belief in us, your unwavering belief

and support in your children, and your prayers and encouragement keep us moving every day. This work is due to your guidance and endless support.

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CHAPTER 1

INTRODUCTION

1.1 Project Context/Background of the Study

In the rapidly advancing arena of educational research, Educational Data Mining (EDM) and Learning Analytics (LA) have proven transformative in enhancing institutional abilities to monitor and predict student performance. Traditional assessment paradigms centered around grades or instructor evaluations often fail to capture the multitude of dimensions affecting academic success. EDM and LA leverage emerging machine learning (ML) capabilities to provide higher education institutions with tools for predictive analytics and decision-making.

Machine learning algorithms such as Random Forest (RF), Support Vector Machines (SVM), Decision Trees (DT), and XGBoost are pivotal in analyzing multi-faceted educational datasets. While Random Forest is highly effective in handling nonlinear relationships, Decision Trees offer interpretative simplicity, especially in datasets with limited complexity. XGBoost, known for iterative refinement, has demonstrated substantial gains in accuracy for predicting academic risks and student dropout rates [1, 2]. Rigorous comparative analyses of ML frameworks reveal that ensemble models—like the Ant Colony Optimization-Decision Tree (ACO-DT) combination and hybrid XGBoost-RF—outperform single algorithms in precision and early detection of at-risk students [3, 4].

Behavioral data integration serves to enhance the robustness of predictive systems. Incorporating metrics such as Learning Management System (LMS) activity, forum engagement, and task-specific time allocations can yield higher prediction accuracy while

fostering adaptive learning environments. Research confirms that combining behavioral data with academic records (e.g., grades, attendance) yields actionable insights for personalized interventions. For example, one study utilized Moodle LMS data to provide real-time alerts for academic advising [5, 6].

Localized research in contexts like the Philippines exemplifies the power of predictive models tailored to institutional environments. The application of Decision Trees to diploma programs demonstrated over 75% accuracy in predicting outcomes correlated with attendance and prior grades [7, 8]. National datasets such as PISA have been instrumental in enabling predictive analytics to address macro-level educational challenges. At the global level, studies from the UK illustrate the role of EDM in supporting individualized learning by creating interventions aligned with student-specific needs [9, 10].

Interoperability in data integration is increasingly recognized as a critical component of successful predictive analytics frameworks. Efficient integration of institutional, national, and behavioral datasets ensures scalability while addressing the diversity of educational settings. Robust frameworks like those leveraging balanced datasets improve both prediction reliability and model equity [11, 12]. For example, handling underrepresented class imbalances within datasets significantly enhances the efficacy of Random Forest models, particularly for smaller-scale applications [5, 6].

The growing integration of ensemble systems into educational data mining reflects a paradigm shift toward scalable, predictive analytics frameworks that balance performance

with interpretability. Personalized interventions are invaluable in not only improving academic outcomes but also shaping equitable, learner-centered experiences. Teacher insights add value by contextualizing predictive findings within practical academic realities, driving innovations in differentiated learning environments [13, 14, 15].

Findings from multi-modal studies suggest that integrating demographic, behavioral, and institutional data enables proactive decision-making for academic support and allocation of resources [16, 17, 18, 19, 20]. As machine learning technologies advance, the capacity to refine predictive frameworks grows, inviting future research into optimizing both algorithms and the educational systems they serve.

This work advances prior research by nominating five machine learning classifiers—Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes—for head-to-head performance comparison using XAI techniques. All student data used are limited to academic, enrollment, course load, library usage, online activity logs, and study habits questionnaire results.

The dataset aggregates records from 1st to 3rd year students enrolled in BSIT, BSCS, and BLIS programs (IT, CS, BLIS Departments), with all selected students identified as face-to-face (F2F) enrollees. The aggregated features include academic grades, course load (number of units), library usage logs, online LMS engagement, study habits questionnaire responses, and other institutional indicators.

1.2 Literature Review

This chapter presents a comprehensive review of current research and technological developments related to predictive analytics in education, specifically focusing on the use of machine learning algorithms such as Random Forest to forecast student academic standing. The discussion explores how Educational Data Mining (EDM) and Learning Analytics (LA) have evolved to support early identification of at-risk students and enable data-driven decision-making in higher education. It further examines the effectiveness of various machine learning approaches in academic prediction, emphasizing the advantages of ensemble models like Random Forest for handling complex and multidimensional datasets. The review is organized thematically into seven major sections: (1) Educational Data Mining and Learning Analytics, (2) Machine Learning and Student Performance Prediction, (3) Application of the Random Forest Algorithm in Education, (4) Feature Selection and Data Preprocessing, (5) Model Evaluation Metrics, (6) Explainable Artificial Intelligence (XAI), and (7) Predictive Analytics in the Philippine Context. The final section synthesizes the findings, identifies existing research gaps, and establishes the foundation for developing a Random Forest–based predictive model tailored to the needs of Saint Michael College of Caraga (SMCC).

EDUCATIONAL DATA MINING AND LEARNING ANALYTICS

In recent years, higher education institutions have increasingly adopted **Educational Data Mining (EDM)** and **Learning Analytics (LA)** to enhance understanding of student performance and engagement. Traditional assessment methods, such as grades or instructor observations,

provide limited insights into student learning behaviors. EDM and LA leverage data from Learning Management Systems (LMS), discussion forums, and assessments to enable data-driven decision-making and proactive interventions [1]–[5].

Aldowah et al. [1] highlight the role of EDM and LA in improving learning outcomes and institutional decision-making. Zheng and Li [3] emphasize the use of EDM in STEM education, predicting student disengagement and optimizing instructional strategies. Daniela and Mirela [2] discuss the integration of student activity logs to monitor learning behaviors, while Pektaş et al. [4] propose machine learning models for detecting misconceptions and personalizing interventions. Angeioplastis et al. [5] focus on leveraging LMS data with predictive algorithms to anticipate student performance.

II. Machine Learning in Student Performance Prediction

Machine learning models are central to predictive analytics in education, with **Random Forest (RF)**, Gradient Boosting, Support Vector Machines (SVM), and Decision Trees among the most widely applied techniques [6]–[10]. These algorithms identify patterns in academic, behavioral, and demographic datasets, enabling early detection of students at risk of poor performance or dropout.

Al-Ameri et al. [6] demonstrate the effectiveness of ensemble models, including RF, in predicting academic outcomes from multimedia data. Sawarkar et al. [7] combine RF with Logistic Regression to improve accuracy in identifying at-risk learners. Borna et al. [8] analyze clickstream data to enhance prediction reliability, while Vijayalakshmi and Srinath [10] examine the impact of collaboration metrics on learning performance. Muhammad et al. [9] highlight personalized education datasets to support adaptive learning strategies.

APPLICATION MACHINE LEARNING CLASSIFICATION ALGORITHM MODELS

Classification algorithms play a fundamental role in data-driven research, with their performance shaped by factors such as the choice of algorithm, preprocessing, and evaluation metrics. This empirical study systematically compares Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes on benchmark datasets, finding that Random Forest consistently demonstrates the highest accuracy, precision, recall, and F1-score. The results indicate that no single evaluation metric is universally optimal; rather, a combination of metrics better captures the varied aspects of classification performance. These findings underscore the importance of careful metric and algorithm selection, while also suggesting the necessity of contextualizing classifier performance to the requirements of specific applications .

FEATURE SELECTION AND DATA PREPROCESSING

Data preprocessing and feature engineering significantly affect model performance. Techniques such as **normalization, SMOTE for class imbalance, and categorical encoding** are widely adopted to improve predictive accuracy [16]–[20].

Angeioplastis et al. [16] emphasize preprocessing to handle imbalanced datasets. Vijayalakshmi and Srinath [17] highlight the integration of demographic and behavioral features. Pektaş et al. [18] and Chauhan and Mishra [19] demonstrate systematic feature engineering to improve

classification outcomes, while Gul et al. [20] discuss preprocessing in the context of deep learning models for academic prediction.

MODEL EVALUATION METRICS

Evaluating predictive models requires standardized metrics to assess accuracy and reliability. Accuracy, precision, recall, and F1-score are frequently used to quantify model performance in educational prediction tasks [6], [21]–[24]. Al-Ameri et al. [6] emphasize the use of balanced metrics to evaluate RF performance, especially in imbalanced datasets. Sawarkar et al. [7] and Madahana & Ekoru [22] demonstrate comparative evaluation with Gradient Boosting and Decision Trees. Angeioplastis et al. [23] provide guidelines for cross-validation and stratified sampling to ensure generalizability, while Borna et al. [24] incorporate probabilistic metrics to assess prediction confidence.

EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

Despite their predictive power, machine learning models are often criticized as "black boxes." **Explainable AI (XAI)** techniques such as SHAP and LIME help educators interpret model predictions and trust intervention strategies [25]–[28].

Borna et al. [25] integrate LIME for visualization of feature importance in student performance prediction. Aldowah et al. [26] propose XAI frameworks in LA systems for transparency. Pektaş et al. [27] and Chauhan & Mishra [28] illustrate how interpretable insights can guide interventions, ensuring fairness and accountability.

VII. Predictive Analytics in the Philippine Context

Localized applications are critical for small-scale institutions like Saint Michael College of Caraga (SMCC). Limited data availability and specific demographic characteristics require customized predictive models [29], [30]. Muhammad et al. [29] provide structured datasets supporting RF-based prediction in Philippine institutions, while Aldowah et al. [30] highlight challenges such as sparse datasets and resource constraints.

1.3 Project Objectives

General Objectives:

To develop a predictive system utilizing machine learning algorithm models for classifying student academic standing, ensuring interpretable results and actionable insights for educational decision-making.

Specific Objectives:

1. To analyze requirements and identify the necessary data sources, system functions, and user needs for the predictive model.
2. To design a functional system that integrates institutional datasets for automated academic standing classification.
3. To evaluate the model's performance using accuracy, precision, recall, F1-score, and ROC-AUC metrics.
4. To assess the system's functionality, performance efficiency, usability, and reliability based on ISO 25010 standards to ensure software quality.

5. Implement and compare multiple machine learning models including Decision Tree, Random Forest, SVM, KNN, and Naive Bayes.
6. Incorporate Explainable Artificial Intelligence (XAI) techniques for model interpretability and transparency.

1.4 Scope and Limitations

This study focuses on the development of a predictive analytics system that employs the Random Forest algorithm to forecast student academic standing at Saint Michael College of Caraga, particularly within the College of Computing and Information Sciences (CCIS). The dataset includes third-year students enrolled in BSIT, BSCS, and BLIS programs. The model will classify students as Good Standing, On Probation, or At-Risk, based on variables such as academic performance, attendance, demographics, and socioeconomic data. The predictive results will be presented through a decision-support dashboard for administrators and faculty to aid in early academic intervention and retention planning.

The model is limited to datasets collected from SMCC, which may affect generalizability to other institutions. Incomplete, inconsistent, or biased data may influence prediction accuracy. The system does not include non-academic factors such as mental health, employment status, or internet access, which also affect performance. Ethical considerations—such as data privacy, bias, and potential misclassification—are recognized, and measures will be taken to ensure confidentiality and responsible use. The prototype does

not yet support long-term tracking or qualitative validation, and results are intended to assist, not replace, academic judgment.

1.5. Definition of Terms

In this study, the following terms are defined operationally based on their application in developing and evaluating a Random Forest-based predictive model for student academic performance at Saint Michael College of Caraga:

Academic Standing - Refers to the classification of students at Saint Michael College of Caraga based on their overall academic performance. In this study, students are categorized as *Good Standing*, *On Probation*, or *At-Risk* according to the predictive output of the Random Forest model.

Accuracy - The share of predictions the Random Forest model gets right out of all predictions made, giving a general sense of how well the classifier identifies students' academic standing overall. While useful as a headline metric, it is best interpreted alongside precision, recall, and the confusion matrix for a more comprehensive performance evaluation.

At-Risk - A classification output from the predictive model indicating students with a high likelihood of academic failure or dropout. These students are prioritized for academic counseling and targeted intervention based on their predicted risk level.

Confusion Matrix - A tabular representation used to evaluate the model's performance by comparing predicted outcomes with actual results. It shows how many instances were correctly and incorrectly classified for each category of academic standing.

Data Cleaning - A crucial step in data preprocessing that involves detecting, correcting, or removing inaccurate, incomplete, or inconsistent data entries from the dataset to ensure high-quality and reliable model training results.

Data Preprocessing - The process of cleaning, transforming, and organizing raw institutional data to prepare it for model training. This includes handling missing values, encoding categorical variables, and balancing class distributions to improve prediction accuracy.

Decision-Support System - The functional component of the study that presents predictive results through an interactive dashboard. It enables faculty and administrators to visualize model outcomes, monitor student performance, and make timely, data-driven academic decisions.

Educational Data Mining (EDM) - The process of applying data mining techniques to educational datasets to uncover hidden patterns and trends. In this study, EDM is used to identify academic, demographic, and socioeconomic factors that affect student performance and academic standing.

Explainable Artificial Intelligence (XAI) - A framework of methods and tools that make the predictions of machine learning models understandable to humans. The study integrates XAI to

help educators interpret the model's reasoning behind each academic standing classification, ensuring transparency and trust.

F1-Score - A model evaluation metric that combines precision and recall into a single measure, representing their harmonic mean. It helps determine the balance between correctly identifying at-risk students and minimizing false classifications.

Feature Selection - The process of determining which input variables contribute most significantly to predicting student academic performance. It enhances model interpretability and reduces computational complexity by eliminating irrelevant or redundant features.

Good Standing - A predictive label assigned to students who meet or exceed institutional academic standards. Students in this category exhibit consistent performance and require minimal intervention from academic advisers.

Learning Analytics (LA) - The measurement and analysis of student learning behaviors using data-driven techniques. Within this project, LA supports the early identification of performance trends and assists in developing strategies for improving academic outcomes.

Machine Learning (ML) - A subset of artificial intelligence that enables computers to learn patterns from data without being explicitly programmed. This study applies ML techniques,

specifically the Random Forest algorithm, to predict students' academic standing based on historical and institutional data.

Precision - A metric that measures the proportion of correctly predicted instances among all instances predicted for a specific category. In this study, it shows how accurately the model identifies students as Good Standing, On Probation, or At-Risk without false positives.

Predictive Analytics - The overarching framework of this study, involving statistical and machine learning methods to forecast student academic standing. It enables the institution to identify potential academic risks and implement early interventions.

Recall - A model evaluation metric that measures the proportion of actual instances of a class (for example, all At-Risk students) that the model correctly identifies. It reflects the model's sensitivity in detecting students who truly require academic intervention.

ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) - A performance metric that evaluates the model's ability to distinguish between different classes. A higher ROC-AUC value indicates a stronger predictive capability in classifying students accurately across all categories.

SMOTE (Synthetic Minority Oversampling Technique) - A data-balancing method used during preprocessing to address class imbalance by generating synthetic examples of underrepresented categories, such as the At-Risk class, to improve model fairness and accuracy.

Student Retention - The institutional objective of keeping students enrolled and progressing toward graduation. The predictive model contributes to this goal by identifying students at risk of academic failure early, allowing for proactive support measures.

CHAPTER 2

2.1 Research Design

This study employs a quantitative, developmental research design. The quantitative approach is essential for the statistical analysis of student data, which includes academic, demographic, and socioeconomic variables. This design allows for the application of machine learning techniques to identify patterns and build a predictive model. The developmental aspect of the research involves the systematic design, creation, and evaluation of the predictive system as a software artifact.

2.2 Software Development Life Cycle



Figure 1: Comprehensive Methodology

*This study employs a **comprehensive methodology** that synergizes the structured phases of an iterative **Software Development Life Cycle (SDLC)** with the cyclical, data-centric phases of the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**. This integrated framework provides a robust structure for managing both the software engineering and data science components of the project. The SDLC governs the overall project architecture, from requirements analysis to final system deployment, while the CRISP-DM framework is nested within it to guide the iterative processes of data understanding, preparation, modeling, and evaluation. This dual approach ensures that the development of the predictive model is both methodologically sound and aligned with the software implementation goals.*

2.3 System Architecture

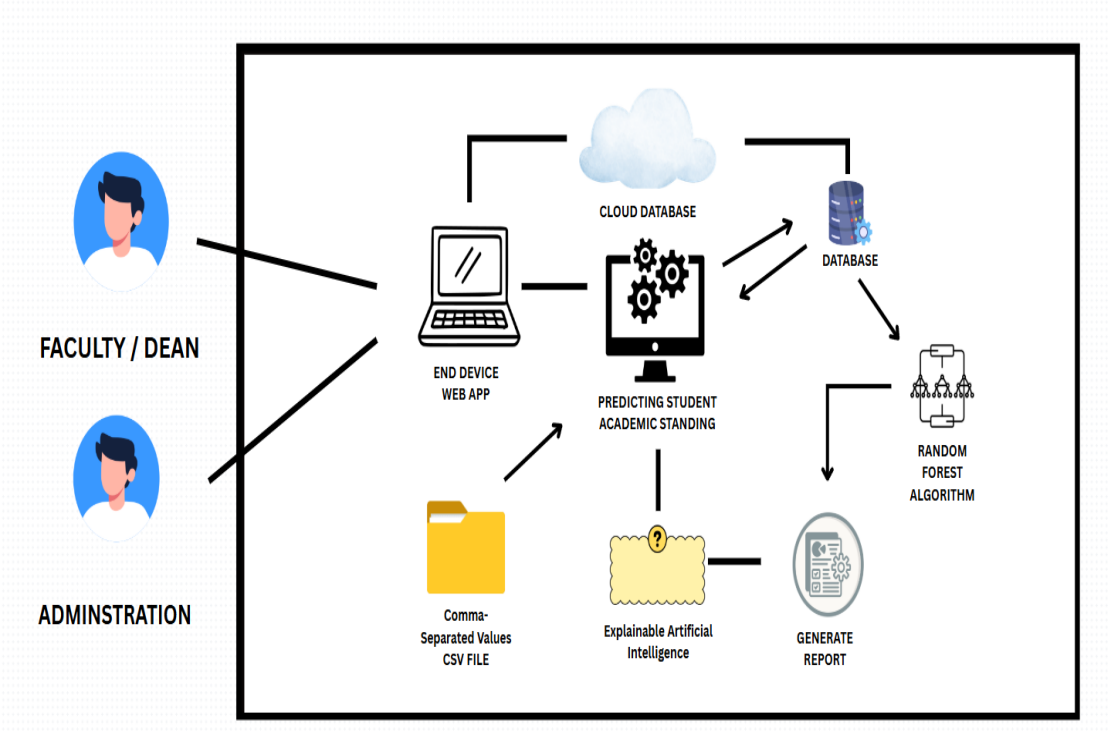


Figure 2:System Architecture

Figure 2: Illustrates the system is designed using a multi-tier web architecture, a standard paradigm for developing robust and scalable web-based applications. This architecture logically separates the system into three distinct layers: the Client-Side, the Server-Side, and the Data Tier. This separation ensures modularity, simplifies maintenance, and enhances security by isolating the business logic and data storage from the user interface.

2.4 Conceptual Framework

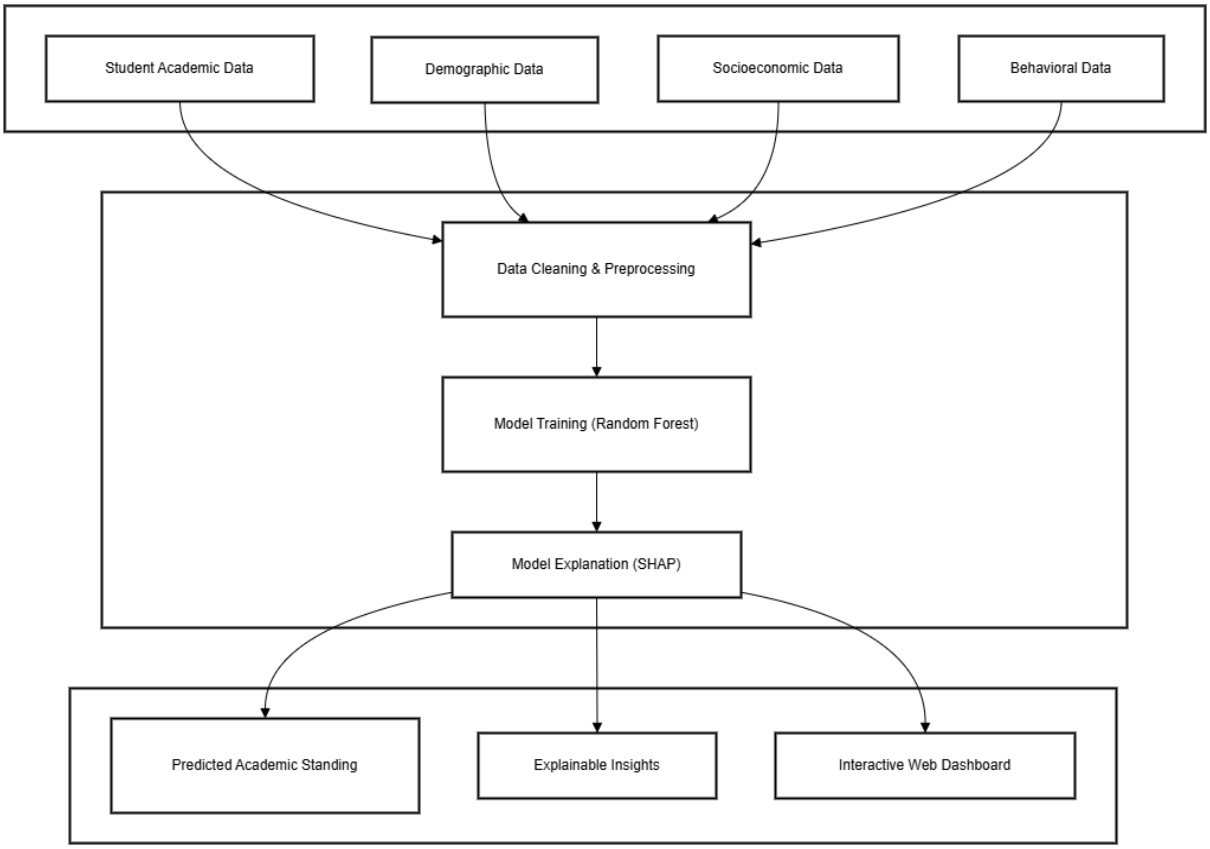


Figure 3: Conceptual Framework

Figure 3 the conceptual framework of this study is illustrated using an Input-Process-Output (IPO) model. This model provides a clear, high-level overview of the research flow, demonstrating how raw data is systematically transformed into actionable intelligence for educational stakeholders. The framework is composed of three sequential stages: the Input, the Process, and the Output.

2.5 Algorithms / Framework Used

Random Forest Algorithm

This study utilizes the Random Forest algorithm as the primary predictive model due to its high accuracy, robustness to overfitting, and ability to handle complex and multidimensional educational datasets. Random Forest operates by constructing an ensemble of decision trees (T_1, T_2, ..., T_m), each trained on bootstrapped samples of the training data with random feature selection at each split to increase diversity.

For regression tasks, the predicted outcome $\hat{y}$ for an input vector x is the mean prediction across all trees:

$$\hat{y} = (1 / M) \Sigma [j=1 \text{ to } M] T_j(x)$$

For classification tasks, the prediction $\hat{C}$ is obtained through majority voting of class predictions from each individual tree:

$$\hat{C} = \text{mode } \{T_1(x), T_2(x), ..., T_m(x)\}$$

Each decision tree recursively partitions the input space by selecting splits that minimize an impurity measure such as the Gini impurity G, calculated as:

$$G = 1 - \Sigma [k=1 \text{ to } K] p_k^2$$

where p_k is the proportion of class k in the node, and K is the total number of classes. This ensemble learning approach improves generalization and predictive stability, making it well-suited for educational prediction tasks.

Explainable Artificial Intelligence (XAI) Framework

Explainable Artificial Intelligence (XAI) is a framework designed to make machine learning models more transparent and interpretable. It encompasses a variety of algorithms, such as SHAP and LIME, that quantify and visualize how individual features influence model predictions. In this study, SHAP is integrated as the primary XAI method for explaining Random Forest outputs.

To enhance interpretability and stakeholder trust, the SHAP (SHapley Additive exPlanations) framework is incorporated to decompose individual model predictions into additive feature contributions ϕ_i. SHAP values are grounded in cooperative game theory, providing a mathematically principled method for attributing the effect of each feature.

The prediction model f(x) for an instance x can be expressed as:

$$f(x) = E[f(X)] + \Sigma\ [i=1 \text{ to } M]\ \phi_i$$

where E[f(X)] is the expected output of the model over the dataset and M is the total number of input features. The SHAP values ϕ_i quantify each feature's marginal contribution averaged over all possible feature combinations. SHAP enables clear, consistent explanations of complex model outputs, supporting actionable insights for targeted academic interventions.

Model Evaluation Metrics

The model's predictive performance is assessed using standard classification metrics defined as follows:

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision assesses the correctness of positive predictions:

Precision = TP / (TP + FP)

Recall (Sensitivity) gauges the completeness of positive predictions:

Recall = TP / (TP + FN)

F1-Score provides a harmonic mean of precision and recall balancing false positives and false negatives:

$F_1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Here, TP, TN, FP, and FN signify true positives, true negatives, false positives, and false negatives, respectively. Additionally, the area under the Receiver Operating Characteristic curve (ROC-AUC) is employed to evaluate the model's discrimination capacity across classification thresholds.

2.6 Use case Diagram

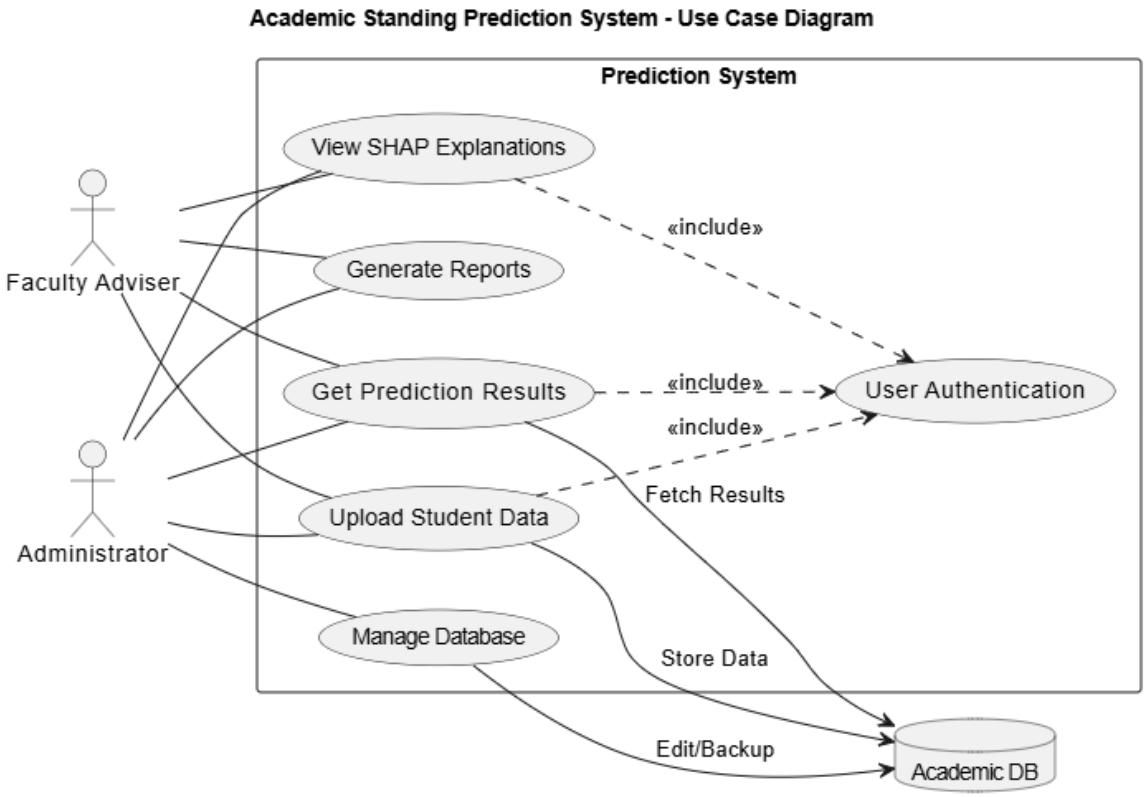


Figure 4: Use case Diagram

Figure 4 illustrates how administrators and faculty advisers interact with the system's core functions, with user authentication securing all critical actions and database interactions ensuring centralized and regulated data management.

2.7 Activity Diagram



Figure 5: Activity Diagram

Figure 5 illustrates how administrators and faculty advisers interact with the prediction system's main workflow, including secure login, data upload, validation, and centralized database operations. After generating prediction results, the system produces SHAP-based explanations, ensuring users receive both outcomes and model reasoning. Decision nodes manage feedback and retries, while swimlanes delineate the responsibilities of users, system, and database throughout the process. The inclusion of SHAP explanations enhances transparency and interpretability for all end users.

2.8 Sequence Diagram

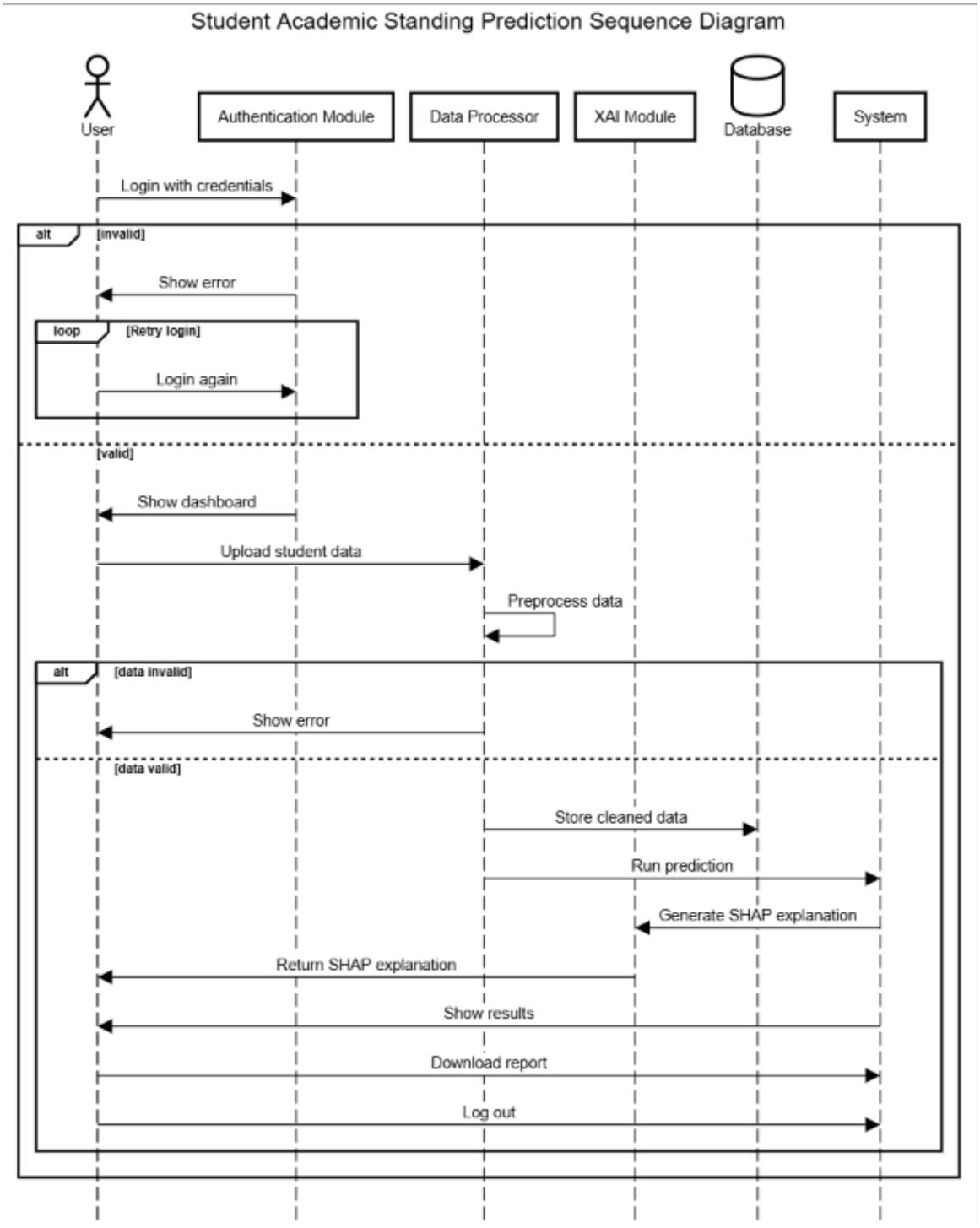


Figure 6: Sequence Diagram

Figure 6 shows how users interact with the prediction system to log in, upload and validate student data, and store it in the centralized database. The system then generates prediction results along with SHAP explanations, providing both outcomes and model reasoning. Conditional branches handle errors and retries, while distinct lifelines represent the roles of users, system modules, and the database. SHAP explanations ensure transparency and interpretability of the predictions.

2.9 ERD (Database Design)

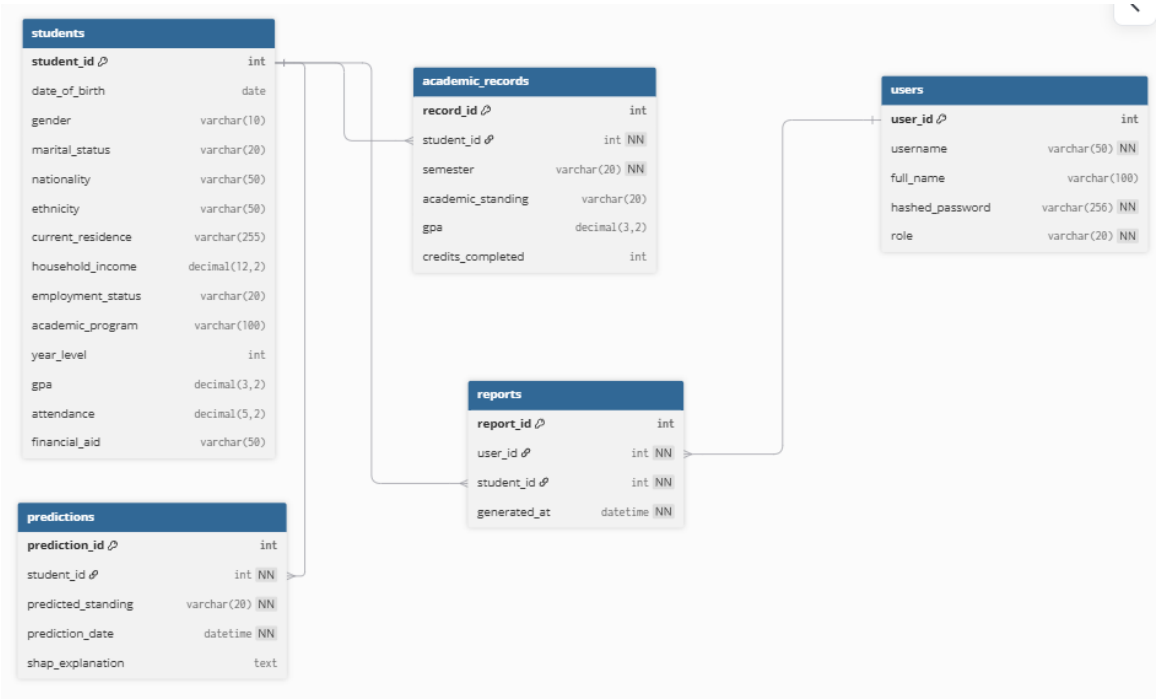


Figure 7: Entity Relationship Diagram

Figure 7 shows the core tables and relationships for managing students, academic records, predictions, user accounts, and reports. Each student is linked to their records and prediction

results, while users can generate reports. This design ensures organized, secure data access for prediction and reporting in the academic system.

2.10 User Interface Design (Prototype)

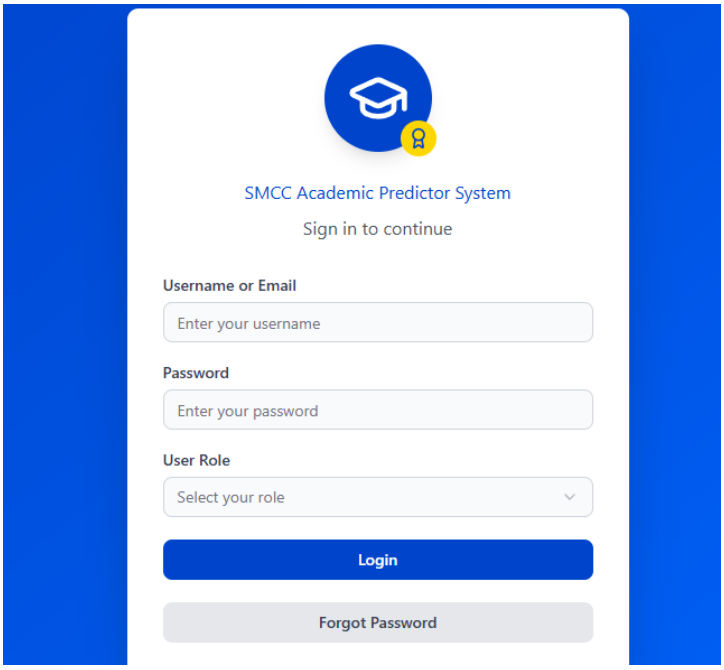


Figure 8: Login Screen

Figure 8 shows how users interact with the prediction system to log in, select their role, and authenticate access to the centralized application. The login form establishes user identity as the gateway to uploading data, generating predictions, and viewing model explanations. Conditional branches support error handling for invalid credentials, while clear lifelines distinguish authorized user roles and secure access, ensuring interpretability and integrity for all subsequent processes.

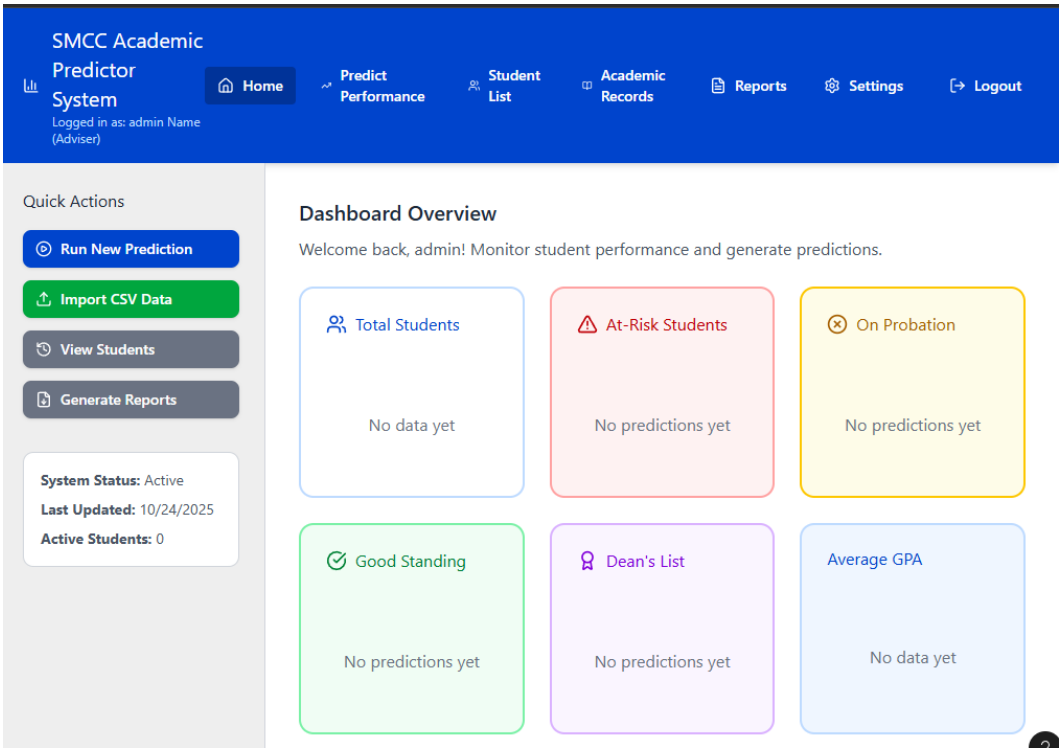


Figure 9: Dashboard/Home

This figure 9 shows how users interact with the prediction system dashboard to view summary metrics, monitor prediction outcomes, and access core features such as running new predictions or importing CSV data. Each dashboard card and quick action links users to data upload, model execution, and result validation. Conditional branches handle empty-state messaging and errors, while key indicators and SHAP-driven output provide a transparent, interpretable overview of prediction results.

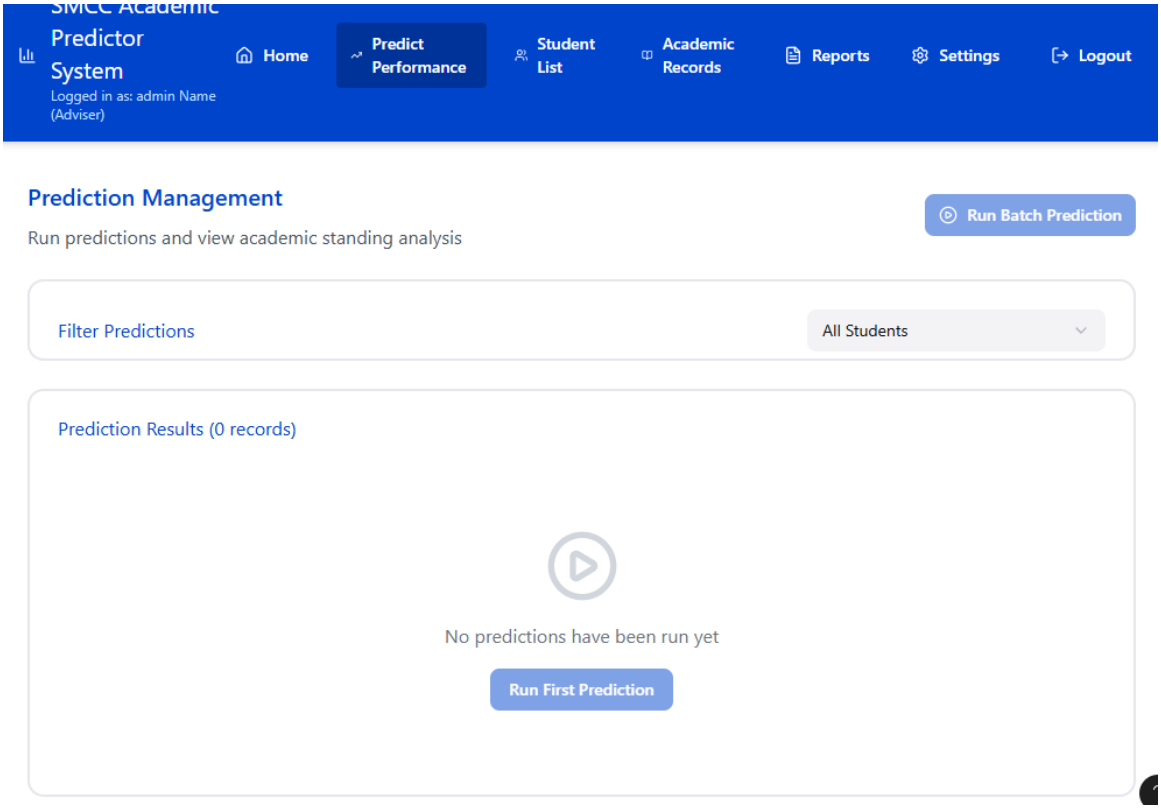


Figure 10: Prediction Management

Figure 10 shows how users interact with the prediction system to manage and execute academic standing predictions. Users can run batch predictions, filter by students, and review prediction results alongside SHAP explanations. The system ensures that both outcomes and reasoning are displayed, supporting transparency and model interpretability. Conditional branches facilitate retries on failures, while distinct interface lifelines support streamlined prediction tasks and XAI visualization.

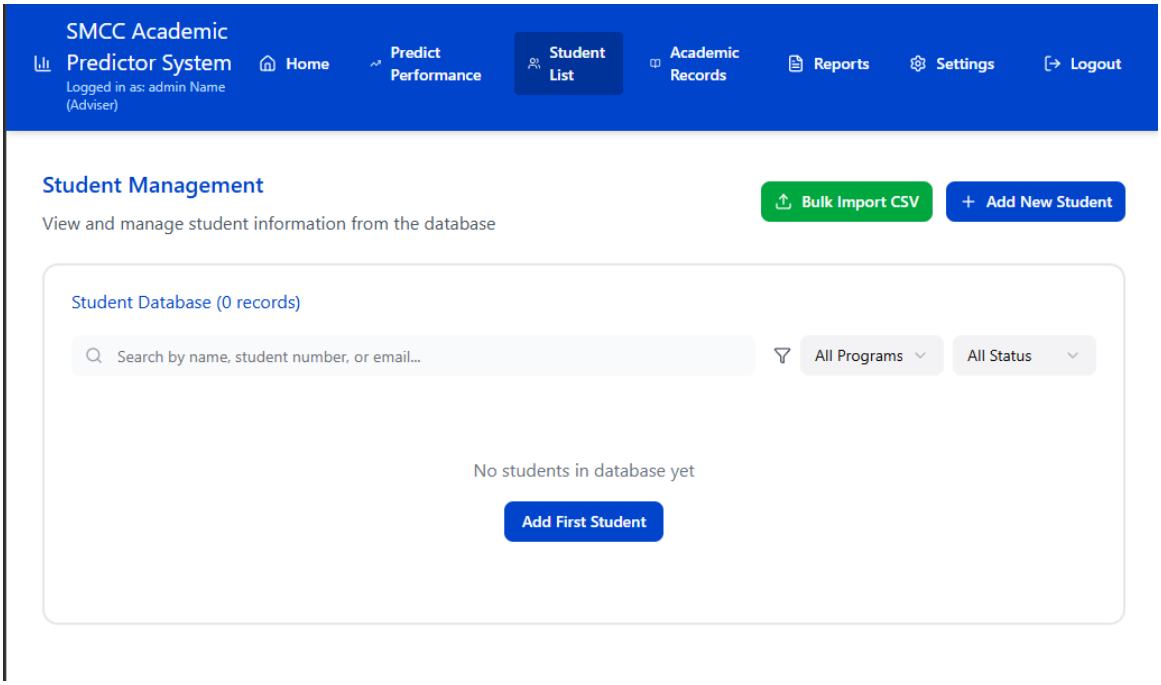


Figure 10: Student Management

This figure shows how users interact with the system to manage the student database, including adding new students and bulk importing via CSV. The management module supports search, filters, and status updates, integrating directly with the centralized database for validated storage. The workflow includes error handling and retry logic, and connects to prediction modules where SHAP explanations ensure interpretability for each student's predicted standing.

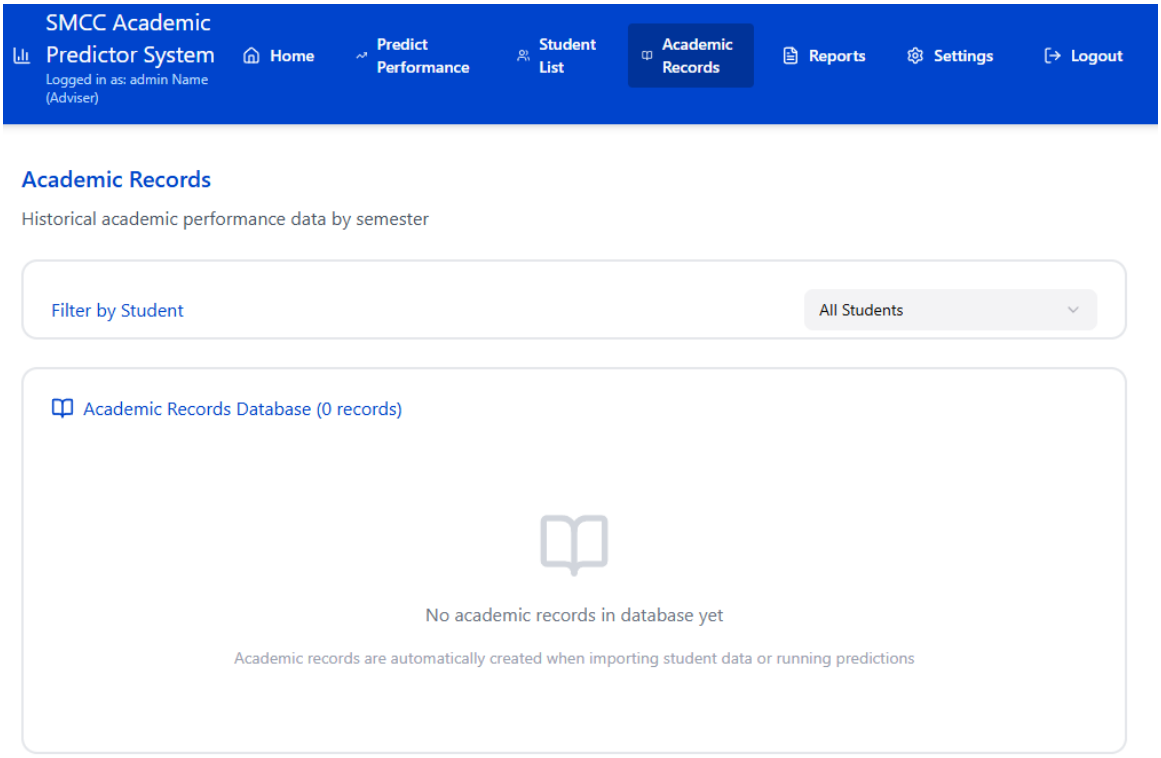


Figure 12: Academic Records

This figure shows how users interact with the prediction system to filter, review, and manage historical academic records by student and semester. Academic records are synchronized with the centralized database and automatically updated by data import and prediction activity. The interface supports conditional messaging for empty data states, while integrating with prediction and SHAP explanation workflows to maintain full transparency and interpretability of academic analysis.

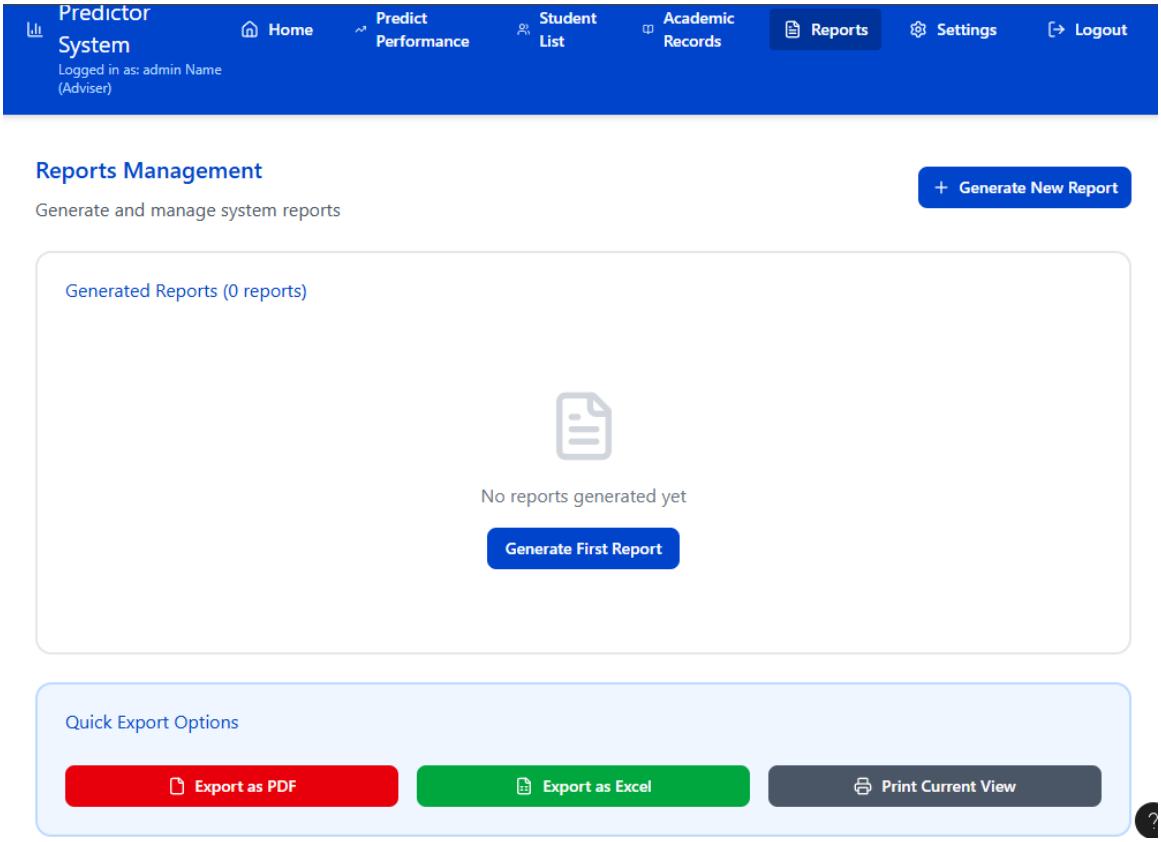


Figure 10: Reports Management

This figure shows how users interact with the prediction system to manage generated reports, export data, and print system views. The reporting interface enables users to create, review, or export results, with the system highlighting error states when no reports exist. Lifeline separation distinguishes reporting, database storage, and user operations, with SHAP explanations integrated into report insights for enhanced transparency and interpretability.

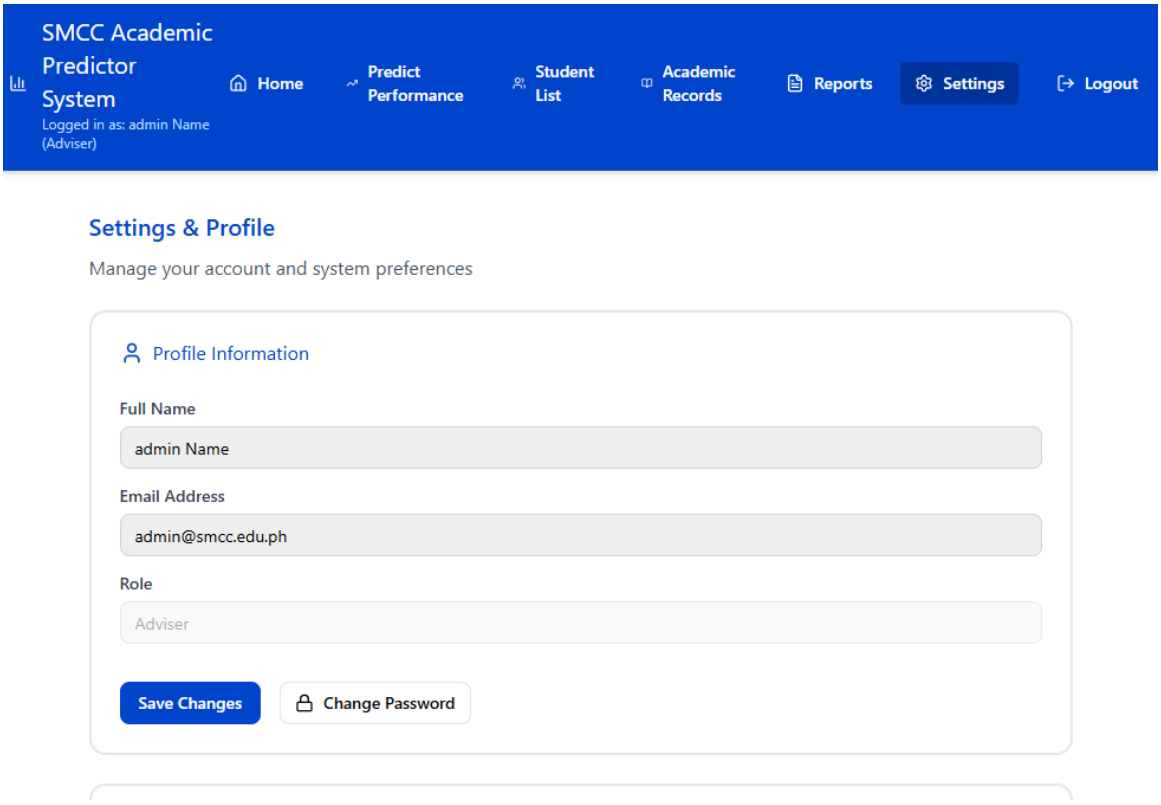


Figure 10: Settings & Profile

This figure shows how users interact with the prediction system to manage their personal account details, update credentials, and review role-based information. Profile management safeguards user authentication and system preferences, with conditional logic handling changes and validation. Distinct system lifelines ensure that user settings are maintained securely and accessed appropriately, supporting traceability and proper data stewardship throughout predictive and explanatory workflows.

2.11 Software Platforms, Development Environments and Tools

Table 1

Software Requirements

Component	Specification	Usage
Front-end		
Jupyter Notebook	**JupyterLab (Anaconda)**	**Interactive Python coding, data visualization, and EDA.**
Streamlit (optional)	**Python Web UI Framework**	**Dashboard for presenting prediction results and user interactions.**
HTML/CSS/JS (optional)	**Web Standards**	**For custom visualization dashboards if web deployment is required.**
Back-end		
Python	**3.8+**	**Develops entire ML pipeline: preprocessing, modeling, automation scripts.**
Flask/FastAPI (optional)	**Python Web Framework**	**Provides REST API endpoints for serving predictions in a web environment.**
scikit-learn	**Python Library**	**Random Forest algorithm, preprocessing, validation, and evaluation.**
pandas	**Python Library**	**Data cleaning, transformation, and merging.**
NumPy	**Python Library**	**Matrix and vector operations for efficient computations.**

SHAP, LIME	**Python Library**	**Model interpretation and feature analysis.**
Matplotlib, Seaborn	**Python Libraries**	**Plots data distributions and evaluation metrics.**
Database/Storage		
MySQL/PostgreSQL	**RDBMS**	**Stores academic, demographic, and LMS records, supports extraction for analysis.**
Excel/CSV	**File Format/Tool**	**Data transfers, backup, and initial cleaning.**
Version Control/Cloud		
Git, GitHub	**Version Control**	**Tracks code development and team collaboration.**
Google Colab/Kaggle	**Cloud Notebooks**	**Cloud computation, easy sharing and scalable model training.**

2.12 Hardware Requirements

Table 2

Hardware Requirements

Components	Specification	Usage
Development PC		
Processor	Intel Core i5 (8th gen) or AMD Ryzen 5	For running data analysis, training models, and development tasks.
RAM	8 GB DDR4	Supports multitasking and fast computation during training and EDA.
Storage	256 GB SSD	Quick access and ample space for large datasets and library dependencies.
Display	1366x768 resolution or higher	Ensures clear visibility of data visualizations and code interfaces.
OS	Windows 10/11, Ubuntu 20.04+, or macOS	Compatibility with necessary ML libraries and tools.
Internet Access	Broadband (10 Mbps+)	Enables data collection, cloud computation, collaboration, and API access.
Peripheral Devices	Mouse, Keyboard, Webcam	Standard input and communication devices for remote collaboration/reporting.

2.13 System Evaluation/Evaluation Procedure

Software Evaluation Using ISO 25010 Standards

The system will be evaluated based on the ISO/IEC 25010 software product quality model, which is the international standard for assessing software quality. ISO 25010 defines eight primary quality characteristics, each with specific sub-characteristics, that will be systematically reviewed during the system evaluation:

- Functional Suitability: Measures how well the system provides functions that meet the stated needs, including completeness, correctness, and appropriateness.
- Reliability: Evaluates stability and error tolerance during operation, including maturity, availability, fault tolerance, and recoverability.
- Performance Efficiency: Assesses system response times, resource utilization, and capacity under specified workloads.
- Usability: Considers the user experience, covering learnability, operability, accessibility, and user interface aesthetics.
- Security: Ensures protection of data and operation against unauthorized access, including confidentiality, integrity, accountability, and non-repudiation.
- Compatibility: Reviews the system's ability to operate in different environments and exchange data with other systems (co-existence and interoperability).

- Maintainability: Examines the ease of modifications, fault diagnosis, and improvements in software (modularity, reusability, analysability, modifiability, testability).
- Portability: Checks adaptability, ease of installation, and replaceability across different environments and platforms.

AI/ML Model Evaluation Metrics

For the predictive analytics component, which uses Random Forest and other machine learning algorithms, system accuracy and robustness will be quantified with the following metrics:

- **Accuracy:** The overall percentage of correctly classified instances among all predictions.
- **Precision**: The proportion of true positive predictions among all positive predictions for a given class.
- **Recall** (Sensitivity): The proportion of actual positives correctly identified by the model.
- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure for imbalanced datasets.
- **ROC-AUC** (Receiver Operating Characteristic - Area Under the Curve): Evaluates the model's ability to distinguish between classes.
- **Confusion Matrix:** Reports the counts of true positives, false positives, true negatives, and false negatives for all categories, supporting granular performance analysis.

By combining the ISO 25010 framework for general software assessment with specific AI evaluation metrics, the system's usability, performance, security, and prediction validity will be ensured, aligning with international best practices for both software engineering and data-driven modeling.

2.13 Ethical Standards

The development and implementation of the SMCC Academic Predictor System strictly follow established ethical standards to ensure data security, participant protection, and research integrity. The system is designed with respect for the rights, privacy, and wellbeing of all involved parties, maintaining transparency and academic rigor throughout the study.

A. Intellectual Property Rights (IPR)

This research utilizes publicly available algorithms and frameworks, appropriately licensed under open-source agreements such as MIT or GPL. All developed code, methodologies, and documentation are the intellectual property of the research team and will be credited accordingly. Proper citations and acknowledgments are provided for adapted or third-party materials to uphold academic integrity and prevent infringement.

B. Informed Consent

All participants involved in the study, including institutional stakeholders and system users, are fully informed of the research objectives, data usage, and their rights to withdraw participation at any time without penalty. Consent is formally obtained, either through written or digital means, prior to data collection or system evaluation.

C. Data Privacy and Confidentiality

Data collected during the research is handled with the utmost confidentiality and stored securely using anonymization, encryption, and access restrictions to prevent unauthorized disclosure. Institutional and participant data will not be shared beyond the scope of research purposes without explicit permission.

D. Voluntary Participation

Participation in system testing, surveys, or research activities is strictly voluntary. Contributors retain the right to discontinue involvement at their discretion, and such decisions will not affect their institutional standing or relationship with the research team.

E. Minimization of Harm

The research team actively minimizes risks of physical, psychological, social, or professional harm to study participants. Testing occurs in controlled environments to avoid disrupting institutional workflows or compromising data security.

F. Beneficence

The project aims to advance institutional capabilities by enhancing academic performance monitoring through ethical, data-driven interventions that benefit students, advisers, and administrators alike.

G. Justice and Fair Participant Selection

All data utilized is selected based on relevance and research goals without bias or discrimination. The predictive system is continually evaluated to ensure fairness and prevent algorithmic discrimination against any group or individual.

H. Data Integrity and Accuracy

High standards of data accuracy and integrity are maintained through systematic data cleaning, validation, and verification processes. Results and system outputs are rigorously audited for correctness and reliability.

I. Transparency in Reporting

The methodology, findings, and limitations are detailed in all reports and user documents. The inclusion of explainable AI techniques such as SHAP enhances transparency, allowing users to understand model decisions clearly.

J. Use of Patented or Copyrighted Materials

The study strictly follows all applicable intellectual property laws. Any patented or copyrighted technologies or materials are used only with appropriate licenses or permissions.

K. Ethical Considerations for Animal and Human Trials

This study does not involve animal experimentation. Human data usage is limited to anonymized educational records collected with fully informed consent, ensuring participant wellbeing throughout.

L. Responsible Use of AI and Related Technologies

AI components are deployed responsibly, with ongoing oversight to mitigate risks of bias, misuse, or harm. System recommendations remain advisory; users are encouraged to apply professional judgment.

M. Ethical Approval

Before system deployment on live data or extensive user testing, ethical approval will be sought and obtained from the appropriate Institutional Review Board (IRB). Documentation of approval and compliance efforts will be retained in project archives.

2.14 Data Gathering Procedure

Data Gathering Procedure

The researchers will implement a systematic and ethical data gathering procedure to ensure the confidentiality and integrity of all student information. The process will be conducted in the following phases:

Phase 1: Securing Consent and Formal Approval

The first step in the data collection process is to obtain informed consent from the student participants. The researchers will distribute a survey that includes a clear and comprehensive consent form, asking for permission to use their academic, personal, and financial data for the study.

Following the collection of student consent, the researchers will draft formal request letters addressed to the heads of the Registrar, and Admission offices of Saint Michael College of Caraga (SMCC). These letters will specify the exact data required for the study. The approval process will follow a formal chain of command:

The letters will first be reviewed and signed by the group's Thesis Adviser.

Next, they will be endorsed by the Research Leader.

Finally, the letters will be submitted to the heads of the respective offices for final approval.

Phase 2: Data Collection and Handling

Upon approval, the researchers will coordinate with each office to collect the necessary data. The requested information includes:

From the Registrar's Office: All academic records and related factors.

From the Admission's Office: Financial information, including monthly family income, sources of income, and details on any financial aid received.

Resources and indicators data reflecting student access and utilization of academic materials. Study habits questionnaires administered through structured interviews focusing on resources utilization and study behaviors.

Course load details, specifically the number of units enrolled per semester.

The data will be provided in a combination of physical paper documents and digital Excel files.

Phase 3: Data Security and Confidentiality

To comply with data privacy regulations and ethical research standards, the researchers will immediately implement a strict data security protocol. Upon receiving the data, the researchers will consult with their Data Adviser. All paper documents will be digitized, and the complete

dataset will be consolidated into password-protected files. Access to these files will be restricted to the researchers only.

For data preservation and to prevent accidental loss, the secured files will be backed up on a private cloud storage service. The collected data will be used exclusively for research purposes and will be permanently and securely deleted from all local and cloud storage locations upon the completion and successful defense of the thesis

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CHAPTER 4

SOFTWARE DEVELOPMENT AND TESTING

4.1 Development Process

4.2 Testing Process

- Accuracy Test

-Functionality Test

-Performance Test

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CHAPTER 5

SUMMARY, CONCLUSION, RECOMMENDATION,

5.1. Summary of Findings

5.2. Conclusion

5.3.Recommendation/s

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Civil Status :

Educational Attainment

Elementary :

Secondary :

Tertiary :

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE

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Weight :

Civil Status :

Educational Attainment

Elementary :

Secondary :

Tertiary :

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

B - Instrument with Informed Consent

C - Map of the Research Locale *(Include Coordinates)*

D - Curriculum Vitae

E - Narrative & Photo Documentation

F- Ethics Committee Approval



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